

FBS Totals Betting System: Exhaustive Model Landscape and Frontier Methods

Research cutoff: September 8, 2026. This report extends the prior FBS totals research report, with expanded emphasis on Section 3 (established and underexplored models).

1. Feature ideas with evidence (condensed update)

No direct FBS totals evidence found for any individual tempo, efficiency, or matchup feature in isolation; all remain "mechanism-only" or "raw-score" evidence. The 2023 NCAA clock-rule change reduced plays per game by roughly 7.8% and game length by about 1.4% in early post-implementation analysis. Wind-threshold total-scoring claims from industry sources remain anecdotal. Returning production and recruiting-talent measures are raw team-strength proxies likely to substantially proxy market-embedded team-strength perception.

Feature group	Scoring mechanism	FBS totals evidence	Incremental-to-market evidence	Historical as-of availability	Leakage risk	Cost	Priority
Tempo/possessions	More snaps -> more scoring chances	No direct FBS totals evidence found	Untested	Yes, if strictly trailing	Low-medium	Medium	High
Efficiency (EPA/success rate)	Point-value estimate per play	No direct FBS totals evidence found	Untested	Yes	Medium	Medium-high	High
Weather (wind/temp)	Passing/kicking disruption	No direct FBS totals evidence found; anecdotal thresholds	Untested	Only via paid forecast archive	High	Medium	Medium
Returning production/recruiting	Preseason roster-quality proxy	No direct FBS totals evidence found	Likely proxies market	Yes, preseason	Medium	Low	Medium
Market total/spread/price	Consensus forecast	Adjacent CFB team-totals bias evidence	Baseline by definition	Depends on field audit	High if close leaks	Low	Critical

2. Market structure in FBS totals (condensed update)

Arscott (2023, Journal of Sports Economics) shows CFB team-totals lines are biased due to censoring at zero points, with a reported exploitable pattern exceeding a 55% win rate net of typical transaction costs -- profitable-after-vig evidence, but specifically for team totals, not confirmed for the combined game total. Paul and Weinbach's earlier CFB/arena-football totals work finds the over is systematically overpriced due to bettor scoring preference. NFL transfer-hypothesis evidence is mixed: Shank (2018) finds significant NFL totals inefficiency including extreme-total bias, while a separate NFL preseason totals study (1996-2019) finds a null result (efficient market, ~50.8% under-cover rate). These conflicting NFL results argue against assuming any universal totals bias transfers cleanly to FBS.

Claim	Evidence	Seasons/sample	Line timing	Vig handled	Out of sample	Execution limits	Verdict
CFB team-totals censoring bias	Peer-reviewed	Not reported (secondary: ~20yrs to 2022/23)	Not reported	Addressed qualitatively	Yes	Not reported	Credible but scope-limited
CFB over systematically overpriced	Peer-reviewed (secondary)	Not reported	Not reported	Not reported	Not reported	Not reported	Directionally suggestive only
NFL extreme-total bias, over-momentum	Peer-reviewed	NFL through ~2018	Not reported	Reported significant	Yes	Not reported	Transfer hypothesis, unconfirmed for CFB
NFL preseason totals efficient (null)	Peer-reviewed	NFL 1996-2019	Not reported	Addressed (50.8%)	Yes	Not reported	Important null/conflicting finding
Wind >13-15mph favors under (CFB)	Industry blog	Unclear N, "since 2005"	Unclear	Not disclosed	Not disclosed	Not disclosed	Anecdotal; unverified

3. Established and underexplored models

Evidence-level framework: (1) mechanism, (2) raw-score forecasting, (3) totals probability, (4) incremental-beyond-market, (5) CLV, (6) profitable-after-vig. Nearly every model family below has zero direct FBS totals evidence at levels 4-6.

3.1 Required baselines

Every model must be compared against: raw market total, calibrated market-only regression, regularized linear scoring model, simple opponent-adjusted rating model, simple residual model, and (for probability methods) market-implied probability baseline. All comparisons must share identical game universe, timestamps, line source, and chronological test window.

3.2 Established model families summary

OLS/ridge/lasso, logistic/ordinal, GAMs, hierarchical/Bayesian partial pooling, dynamic/state-space ratings, random forests, gradient boosting (XGBoost/LightGBM/CatBoost), quantile regression/forests, Poisson/negative-binomial/bivariate models, possession/drive/Monte Carlo simulation, and market-residual/market-as-prior ensembles all have no direct FBS totals evidence found. Poisson/bivariate score models have soccer-analogue adjacent evidence but weaker mechanical fit for football's discrete chunk-scoring. Market-residual ensembles are the priority candidate given direct linkage to Arscott's documented CFB team-totals bias.

3.3 Underexplored probabilistic models

GAMLSS / distributional boosting (NGBoost): Underexplored. No published FBS totals application found (searched: "GAMLSS sports prediction application," "NGBoost sports betting total," cutoff Sept 8 2026). Closest adjacent: MLB Statcast swinging-strike BART modeling; sprint kayak/canoe BART time-correction. Mathematically suited to jointly estimating conditional mean+variance+skew of the total, useful for distinguishing games with same edge but different uncertainty. Prototype: fit NGBoost/GAMLSS on market-residual target, compare CRPS vs. linear baseline. Falsification: if predicted variance uncorrelated with squared residuals out-of-sample, abandon.

BART / heteroskedastic BART: Underexplored. Adjacent evidence: MLB pitch-quality BART; kayak/canoe race-time BART correction. Native Bayesian posterior uncertainty. Prototype: BART vs. linear baseline, compare interval coverage. Falsification: coverage indistinguishable from Normal-approx baseline.

Copula / flexible joint-score models: Underexplored for FBS; directly studied in soccer (bivariate copula goal models, PARX-Copula, CMP-copula HMM for momentum). Football's discrete chunk-scoring is a weaker mechanical fit than soccer's single-goal counting. Prototype: independent marginals + copula fit vs. independent-sum baseline. Falsification: copula dependence parameter statistically indistinguishable from zero.

Normalizing flows / MDN: Speculative. No FBS or clear sports analogue found. Sample-size mismatch (flows typically need more data than ~800-900 games/season) and discrete-score handling issues.

Deep ensembles / Bayesian NN: Poor fit. Sample-size mismatch for base neural net itself makes uncertainty-quantification wrapper premature.

3.4 Underexplored temporal and latent-state models

Dynamic Bayesian latent-team models: Underexplored. Kalman-filter-style weekly updating of offense/defense/pace states, handling roster churn as state innovations rather than rolling-window tuning. Prototype: local-level Kalman filter on one efficiency metric vs. 4-game rolling average, compare one-step-ahead forecast error.

HMM/regime-switching: Underexplored for FBS; adjacent soccer within-match momentum HMM using CMP-copula marginals. Prototype: 2-state HMM vs. known coaching/QB-change dates.

Change-point models (BOCPD): Underexplored. No FBS application found. Same validation logic as HMM: check if detected change points correlate with real personnel/coaching events.

Sequence models (RNN/transformer): Poor fit to speculative given ~12-15 games/team/season; soccer T-GNN work uses much more granular event-stream data, a weak structural analogue.

3.5 Underexplored event-process and structural models

Marked point processes (Hawkes): Underexplored, with important self-excitation caveat -- football's discrete alternating-possession structure differs fundamentally from continuous-time settings where Hawkes excels; sign/magnitude of self-excitation should be treated as empirical, not assumed positive. Falsification: excitation parameter statistically indistinguishable from zero vs. plain renewal/Poisson process.

Survival/competing-risk drive models: Underexplored. Models drive terminal outcome (TD/FG/turnover/punt) as competing risks. Falsification: simulated total mean not better than direct-regression baseline.

Semi-Markov possession models: Underexplored, highest engineering cost in report; most mechanistically faithful representation of possession dynamics.

3.6 Underexplored relational and representation models

Matrix factorization: Underexplored. Latent stylistic offense/defense factors beyond scalar rating. Falsification: rank>1 factorization doesn't beat rank-1 (=scalar rating) on held-out prediction.

Graph neural networks (temporal): Underexplored for pregame FBS; adjacent soccer player-interaction GNN and T-GNN work (different granularity). Requires strictly temporal-causal graph construction (no future-game edges influencing past embeddings). Falsification: no gain over simple SRS-style rating in sparse/early-season prediction.

Multi-task learning: Underexplored. Tests regularization-via-shared-tasks vs. negative transfer. Falsification: total-task accuracy not improved vs. single-task model.

Player/coach embeddings: Underexplored to poor fit -- severe cold-start for players given FBS roster turnover and no comprehensive historical depth-chart archive; coach-only embeddings more feasible.

3.7 Underexplored tabular and automated models

TabPFN: Underexplored. Nature paper reports strong performance on datasets up to 10,000 samples/500 features; TabPFN-2.5 claims scalability to 50,000 samples matching/beating tuned XGBoost. Independent evaluation notes real limitations at high-dimensional/large-scale tasks. CPU-only inference limited to a few thousand samples without GPU/hosted inference. Structurally well-matched to FBS's small-medium data regime, but in-context/transductive design creates a new leakage-risk category: must verify only chronologically-prior games are ever passed as context at each decision point. Falsification: accuracy statistically indistinguishable from XGBoost.

Symbolic regression/genetic programming: Underexplored. Tests whether possessions x efficiency-style formulas can be automatically rediscovered. Acceptance criterion should be stability across chronological folds, not single-fold fit.

3.8 Distribution-shift and robust-learning models

Underexplored across the family (DRO, IRM, Group DRO, meta-learning, conformal under shift). Most tractable entry point: simple era-segmented/weighted training around the 2023 clock-rule break and transfer-portal-era roster churn, plus conformal prediction for interval calibration (with honestly weak formal guarantees under unknown-magnitude shift).

3.9 Market as noisy sensor

Underexplored; data prerequisite (timestamped multi-book quotes) only confirmed available from mid-2020 forward via The Odds API. Industry commentary on sharp-vs-public line movement is anecdotal, undocumented. Critical distinction: predicting the closing total is a fundamentally different task from predicting the final game outcome -- a good closing-line predictor could have zero edge on the actual game.

3.10 Selective prediction and betting decisions

Meta-labeling, conformal risk control, and price-aware EV rules are underexplored for FBS but low-cost and directly relevant. Meta-labeling operationalizes "when to trust the forecast" via a secondary classifier. Reinforcement learning/contextual bandits remain a weak fit given ~800 one-shot annual decisions and no reliable counterfactual feedback loop.

Frontier model summary table

Model	FBS totals use found?	Priority
GAMLSS/NGBoost	No	Priority experiment
BART/heteroskedastic BART	No	Secondary candidate
Copula joint-score models	No	Secondary candidate
Normalizing flows/MDN	No	Weak fit
Deep ensembles/Bayesian NN	No	Poor fit
Dynamic Bayesian latent-state ratings	No	Priority experiment
HMM/regime-switching	No	Conditional candidate
Change-point (BOCPD)	No	Conditional candidate
Sequence models (RNN/transformer)	No	Weak fit
Marked point processes (Hawkes)	No	Conditional candidate
Survival/competing-risk drive models	No	Secondary candidate
Semi-Markov possession models	No	Conditional candidate
Matrix factorization	No	Secondary candidate

Model	FBS totals use found?	Priority
Graph neural networks	No	Secondary candidate
Multi-task learning	No	Conditional candidate
Player/coach embeddings	No	Weak fit (player), Conditional (coach)
TabPFN	No	Priority experiment
Symbolic regression	No	Secondary candidate
Distribution-shift-robust methods	No	Priority experiment (simple version)
Market-as-sensor (Kalman/VAR across books)	No	Conditional candidate
Meta-labeling/selective prediction	No	Priority experiment
RL/contextual bandits for staking	No	Not recommended

4. Validation and leakage (reinforced)

All prior leakage categories apply. Frontier-specific additions: graph/embedding models require strictly temporal-causal graph construction; TabPFN's in-context design requires a dedicated unit test verifying no future game ever enters the context set; distributional models' calibration checks must be performed only on genuinely held-out folds. Nested chronological design (expanding/rolling train window, inner validation, frozen outer test season, weekly retraining, untouched final holdout) applies without exception. A research registry logging every attempted model/feature/threshold combination is recommended.

5. Free or cheap data beyond CFBD

Unchanged from prior report: CFBD (core), The Odds API (multi-book historical odds, NCAAAF from mid-2020 only), Visual Crossing (observed weather free tier; historical-forecast paid tier ~\$165/mo), Meteostat (free observed weather), ESPN returning production (preseason snapshot), recruiting ratings, transfer portal trackers (short history, inconsistent). No FBS historical injury/depth-chart archive identified; prospective collection required going forward. CFBD's own line-field metadata (sportsbook identity, opener/close designation, timestamp granularity) remains unverified at the field level and must be audited first.

6. Ranked experiments

Rank	Experiment	Type	Hypothesis	Data burden	Evidence
1	CFBD line-field audit	Established (diagnostic)	Stored fields reliably represent claimed sportsbook/timing	Low	Foundational, self-verifying
2	Market-only calibration regression	Established	Market shows stable calibration bias in some subgroup	Low	Adjacent CFB team-totals bias

Rank	Experiment	Type	Hypothesis	Data burden	Evidence
3	Market-plus-ML residual hybrid	Established	Engineered features add residual signal beyond market	Medium	No direct evidence found
4	Garbage-time-adjusted possession/drive simulation	Structural	Explicit possession mechanics beat aggregate regression	High	Mechanism-only
5	GAMLSS/NGBoost distributional model	Frontier	Conditional variance/skew improves calibration/filtering	Medium	No direct evidence found
6	Dynamic Bayesian latent-state ratings	Frontier	Kalman updating handles roster churn better than rolling windows	Medium	No direct evidence found
7	Meta-labeling/selective prediction	Frontier	Secondary "trust model" beats fixed edge threshold	Low-medium	No direct evidence found
8	TabPFN walk-forward prototype	Frontier	Foundation-model learner matches/beats boosting on small data	Low-medium	No direct evidence found
9	Multi-book quote-fusion/lead-lag (2020+)	Frontier	Cross-book dynamics predict closing direction/stale prices	Medium-high	Anecdotal only
10	Opener-vs-later-market timing comparison	Established (execution)	Later entry produces different realized CLV/ROI than opener	Medium	No direct evidence; general CLV heuristic (anecdotal)

Immediate roadmap (top 5)

1. CFBD line-field audit -- pass criterion: discrepancy rate below predeclared threshold vs. Odds API 2020+ overlap.
2. Market-only calibration regression -- pass criterion: stable non-zero bias across ≥ 2 independent chronological folds.
3. Market-plus-ML residual hybrid -- pass criterion: residual RMSE improvement holds across ≥ 2 outer seasons plus positive average CLV.
4. Meta-labeling/selective prediction layer -- pass criterion: ROI improvement vs. fixed threshold, stable across seasons.
5. GAMLSS/NGBoost distributional model -- pass criterion: meaningful CRPS improvement plus correct interval coverage.

First experiment to run: CFBD line-field audit

Every downstream experiment depends on trusting stored opener/close fields. Uses only existing data plus a cross-check against The Odds API's 2020+ timestamped history. If fields prove unreliable, the result still redirects effort productively (restrict usable seasons/fields) rather than wasting months on an untrustworthy foundation. Deliverable: a DuckDB audit

report quantifying discrepancy rate, missingness, and staleness, with an explicit go/no-go decision, with the acceptance threshold written down before running the comparison.

Frontier research shortlist (full prototypes)

- 1. GAMLSS/NGBoost distributional regression:** Model residual $y = T_{\text{final}} - T_{\text{market}}$ as $\text{Normal}(\mu(x), \sigma(x)^2)$ with both parameters as additive functions of features, trained via NGBoost's natural-gradient boosting or R gamlss via rpy2. Packages: ngboost, scikit-learn, duckdb. Train on proper scoring rule (NLL/CRPS). Calibrate via PIT/reliability diagrams on held-out folds. Ablate against constant-variance linear baseline. Fail if CRPS/coverage not meaningfully better.
- 2. TabPFN walk-forward prototype:** In-context transformer-based predictor; "training set" = chronologically-prior games only, rebuilt fresh at each weekly decision point. Packages: tabpfn, pandas, duckdb. Must include a unit test verifying no future game appears in context. Calibrate via post-hoc isotonic recalibration. Ablate against XGBoost and linear baseline. Fail if accuracy indistinguishable from XGBoost.
- 3. Meta-labeling/selective prediction layer:** Secondary binary classifier $g(z)$ trained on primary model's historical edge/confidence features to predict whether a bet would be profitable; only bets above threshold are placed. Packages: scikit-learn or lightgbm. Train with profit-weighted loss. Nested walk-forward (meta-model trained only on strictly prior primary-model outputs). Fail if realized ROI not improved vs. fixed point-edge threshold.

Search audit

Search tool: general web search aggregator. Cutoff: September 8, 2026. Every frontier family (GAMLSS, BART, copulas, normalizing flows, dynamic latent-state, HMM, change-point, Hawkes processes, matrix factorization, graph neural networks, multi-task learning, TabPFN) produced "no published FBS totals application found in searched sources" -- reported as a search outcome, not proof of universal absence.

Key references

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